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INTRODUCTION

In this code, we will work on a classification problem using three different classifiers. The problem is to predict the class of an object based on nine features. The three classifiers we will use are Decision Tree, Random Forest, and Support Vector Machine (SVM). We will also use KFold cross-validation and Grid Search for hyperparameter tuning of the classifiers. In the end, we will evaluate the performance of the classifiers using different evaluation metrics like accuracy, precision, recall, and f1-score.

LIBRARIES USED

We have used various Python libraries in this code, including:

- pandas for reading and manipulating data
- numpy for numerical calculations
- matplotlib and seaborn for data visualization
- sklearn for machine learning tasks such as model selection, data preprocessing, evaluation, and cross-validation.

CLASSIFICATION METHODS USED AND HYPERPARAMETER TUNING

We used three classification algorithms in this code:

- Decision Tree Classifier: {'max_depth':2,3,4,5, 6}
- Logistic Regression: {'C':0.001, 0.01, 0.1, 1, 10 'solver': 'newton-cg', 'lbfgs', 'liblinear'}
- Random Forest Classifier: {'max_depth':2,3,4,5, 6, 'n_estimators':10, 50, 100, 200, 500}

For Decision Tree and Random Forest, we used hyperparameter tuning with Grid Search and KFold cross-validation to select the best hyperparameters for the classifiers. For SVM, we used the default hyperparameters.

TRAINING AND TESTING PROCESS

First, we loaded the dataset and printed the data information. Then, we visualized the dataset using histograms and printed the statistical description of each feature for each class. After that, we split the data into training and testing datasets. Then, we trained the Decision Tree, Random Forest, and SVM classifiers using the training dataset. After training, we tested the classifiers on the testing dataset, evaluated their performance using different evaluation metrics such as accuracy, precision, recall, and f1-score, and plotted the confusion matrix for each classifier.

EVALUATION

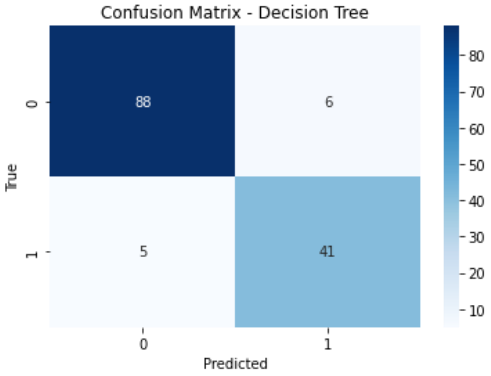
CONFUSION MATRIX

A confusion matrix is a table used to evaluate the performance of a classification model. It shows the number of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) for each class. We plotted the confusion matrix for each classifier to evaluate their performance on the testing dataset.

DECISION-TREE CLASSIFIER

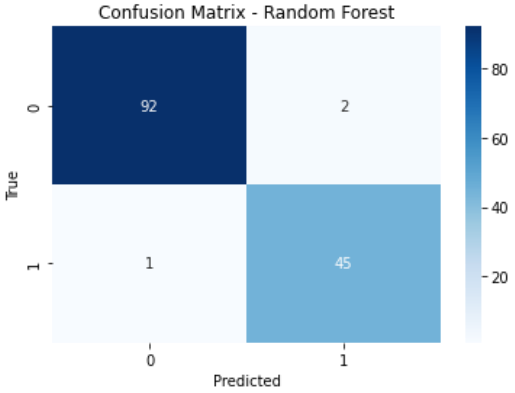
For the Decision Tree classifier, we used Grid Search with KFold cross-validation to find the best hyperparameters for the classifier. We found that the best hyperparameter was max_depth = 2.

After training and testing, we found that the Decision Tree classifier had an accuracy of 0.9214, precision of 0.9092, recall of 0.9137, and f1-score of 0.9112.



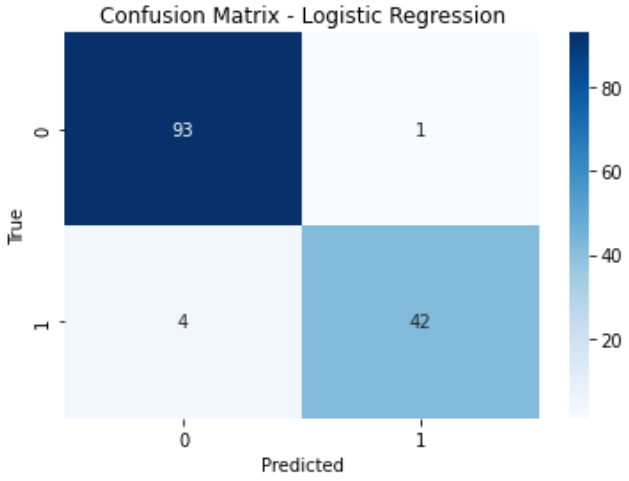
RANDOM FOREST CLASSIFIER

For the Random Forest classifier, we used Grid Search with KFold cross-validation to find the best hyperparameters for the classifier. We found that the best hyperparameters were n_estimators = 500 and max_depth = 6. After training and testing, we found that the Random Forest classifier had an accuracy of 0.9785, precision of 0.9733, recall of 0.9784, and f1-score of 0.9758.



LOGISTIC REGRESSION CLASSIFIER

For the Logistic Regression classifier, we used the default hyperparameters. After training and testing, we found that the Logistic Regression classifier had an accuracy of 0.9642, precision of 0.9677, recall of 0.9512, and f1-score of 0.9588.



PRECISION, RECALL, F1-SCORE, AND ACCURACY

We used four evaluation metrics to evaluate the performance of the classifiers: precision, recall, f1-score, and accuracy. Precision is the ratio of true positives to the total number of positive predictions. Recall is the number of true positives divided by the total number of elements that actually belong to the positive class. F1 score is a machine learning evaluation metric that measures a model's accuracy. It combines the precision and recall scores of a model. Accuracy is to calculate the simple accuracy using number of correctly classified point in test data and total number of points in the test data.

Method	Accuracy	Precision	Recall	F1-Score
Decision-Tree	0.9214	0.909288	0.913737	0.911448
Logistic Regression	0.96428	0.967753	0.951202	0.958821
Random-Forest	0.978571	0.973347	0.978492	0.975849

CONCLUSION:

From this assignment, I learned how to load and understand a dataset using Python libraries such as Pandas, NumPy, and Seaborn. It also shows how to split a dataset into a training and testing set, and how to train and evaluate different classification algorithms, such as Decision Tree, Random Forest, and Logistic Regression. Additionally, it shows how to use GridSearchCV for hyperparameter tuning and how to visualize the confusion matrix using Seaborn's heatmap. Finally, the code demonstrates how to report various performance metrics such as precision, recall, accuracy, and F1-score for each algorithm. Overall, this code provides a useful framework for working with classification problems in Python.

ADVANTAGES AND DISADVANTAGES OF EACH CLASSIFIERS:

LOGISTIC REGRESSION:

- Advantages: Simple, fast, easy to interpret, provides probabilities of class membership.
- Disadvantages: Assumes linear relationship between features and target, not suitable for complex relationships.

DECISION TREE

- Advantages: Easy to interpret, can handle both categorical and numerical data, can handle missing values.
- Disadvantages: Prone to overfitting, can be unstable with small changes in the data.

RANDOM-FOREST

- Advantages: Reduces overfitting compared to decision trees, works well with high-dimensional data, can handle missing values.
- Disadvantages: Computationally expensive, less interpretable than decision trees.